AnalystLab Africa
Data Science Internship — Batch B

Week 3: Statistics & Probability

Summary Report

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Dataset	Rows	Columns	Target Variable
Titanic	891	12 (raw)	Survived (binary)
California Housing	20,640	10 (raw)	median_house_value (continuous)

Objectives for Week 3

1. Descriptive statistics (mean, variance, std dev, skewness, kurtosis)
2. Probability distributions (normal, Bernoulli, Poisson, log-normal)
3. Hypothesis testing (t-test, Chi-square, ANOVA, Mann-Whitney, Kruskal-Wallis)
4. Correlation vs Causation (Pearson, Spearman, effect sizes)

TASK 1: TITANIC DATASET

Dataset: 891 passengers, 12 columns. Target: Survived (0/1). Cleaned: Age imputed (median), Cabin dropped, Embarked imputed (mode). Engineered: FamilySize, IsAlone, FareLog (log1p).

1.1 Descriptive Statistics

Feature	Mean	Std Dev	Median	Skewness	Kurtosis
Age	29.36	13.02	28.00	0.51	0.99
Fare	32.20	49.69	14.45	4.79	33.40
log(Fare)	2.96	0.97	2.74	0.40	0.98
FamilySize	1.91	1.61	1.00	2.73	9.16

Key observations:

- **Fare is heavily right-skewed** (skew=4.79, kurtosis=33.4) — a small number of passengers paid premium fares (max £512), pulling the mean far above the median. A log-transform reduces skewness to 0.40.
- **Age** is approximately normally distributed (skew=0.51). Survivors were slightly younger on average (28.3 vs 30.0 years).
- **1st-class passengers** paid a median fare of £60 — nearly 7.5× the 3rd-class median of £8.

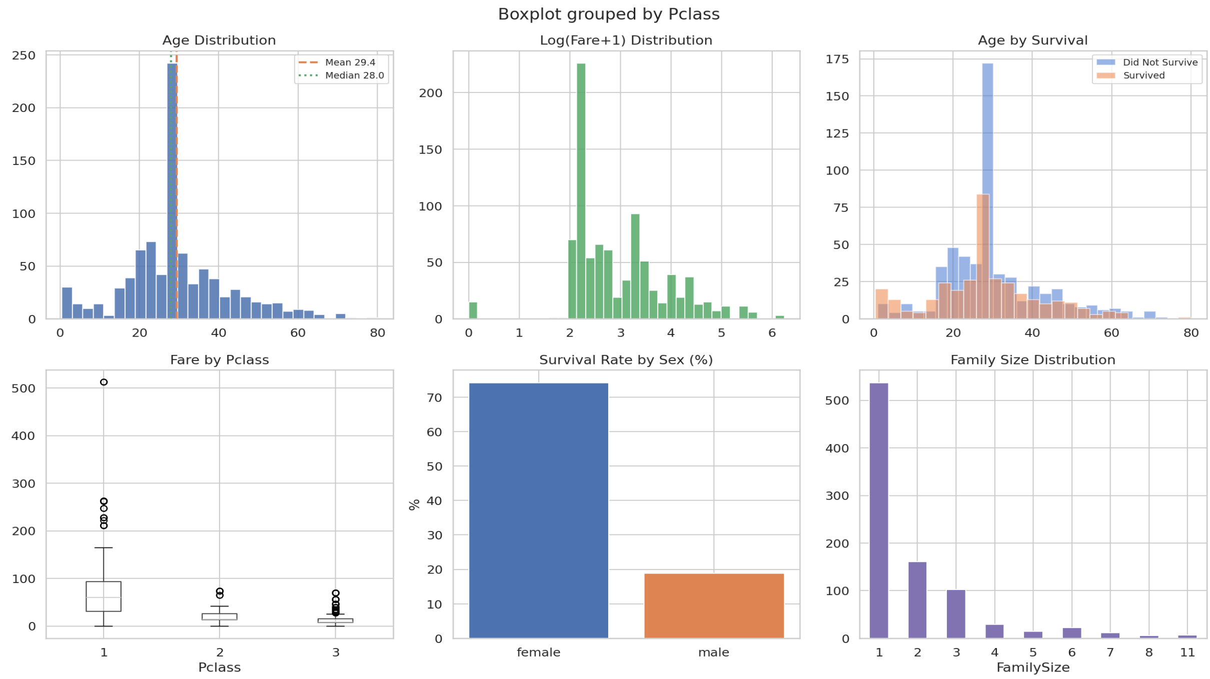

Figure 1.1 — Titanic Descriptive Statistics Overview

1.2 Probability Distributions

Survival as a Bernoulli process:

P(Survival) = 342 / 891 = **0.3838**. The expected number of survivors under a Binomial(891, 0.3838) distribution is exactly 342, matching the observed count.

Conditional survival probabilities:

Group	P(Survive | group)

Female	0.742
Male	0.189
1st Class	0.630
2nd Class	0.473
3rd Class	0.242

Normality: All three (Age, Fare, log(Fare)) reject normality under Shapiro-Wilk (p<0.001). Age is nearest to normal; Fare is highly non-normal; log(Fare) is substantially improved but still not perfectly normal at n=891.

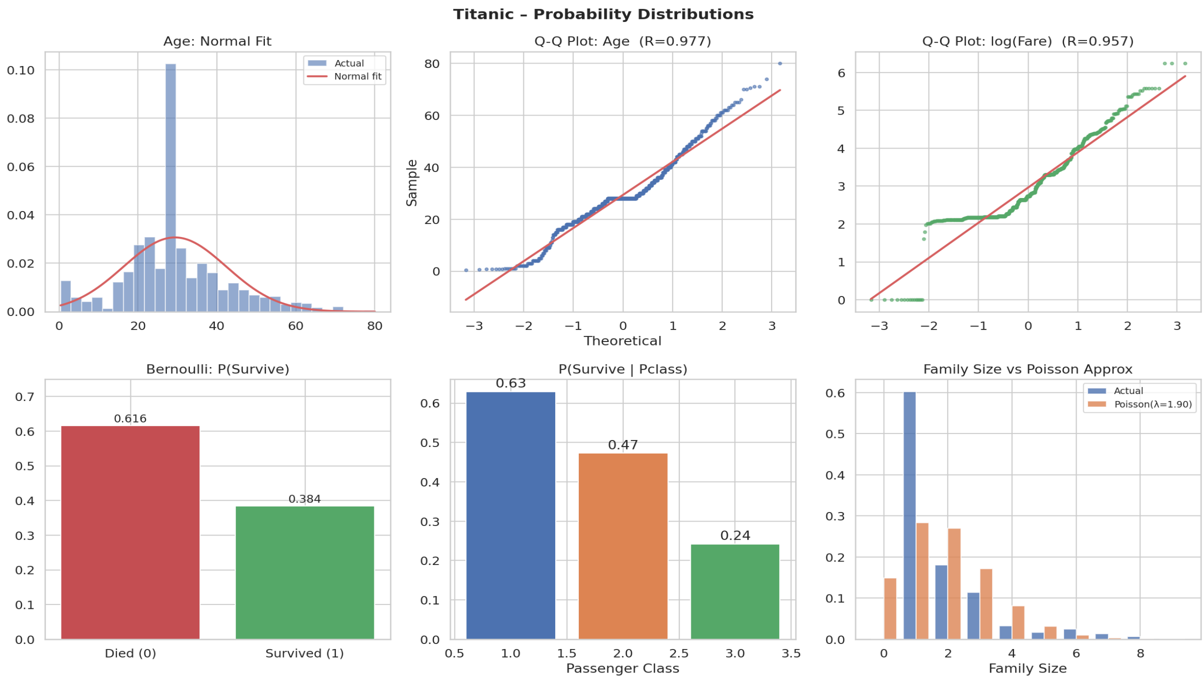

Figure 1.2 — Probability Distributions & Q-Q Plots

1.3 Hypothesis Testing (alpha = 0.05)

Hypothesis	Test	Statistic	p-value	Decision	Effect
H1: Age differs (Survived vs Died)	Welch t	t=–1.90	0.058	Fail to Reject	d=–0.13 (small)
H1b: Age (non-parametric)	Mann-Whitney	U=89,780	0.270	Fail to Reject	—
H2: Sex independent of Survival	Chi-square	X2=260.7	<0.001	Reject H0	V=0.54 (large)
H3: Pclass independent of Survival	Chi-square	X2=102.9	<0.001	Reject H0	—
H4: log(Fare) differs	Welch t	t=10.12	<0.001	Reject H0	—
H5: Embarkation independent	Chi-square	X2=26.5	<0.001	Reject H0	—

Noteworthy: Age alone does not predict survival (both t-test p=0.058 and Mann-Whitney p=0.270 fail to reject H0). Sex is the dominant factor (Cramer's V=0.54, large effect).

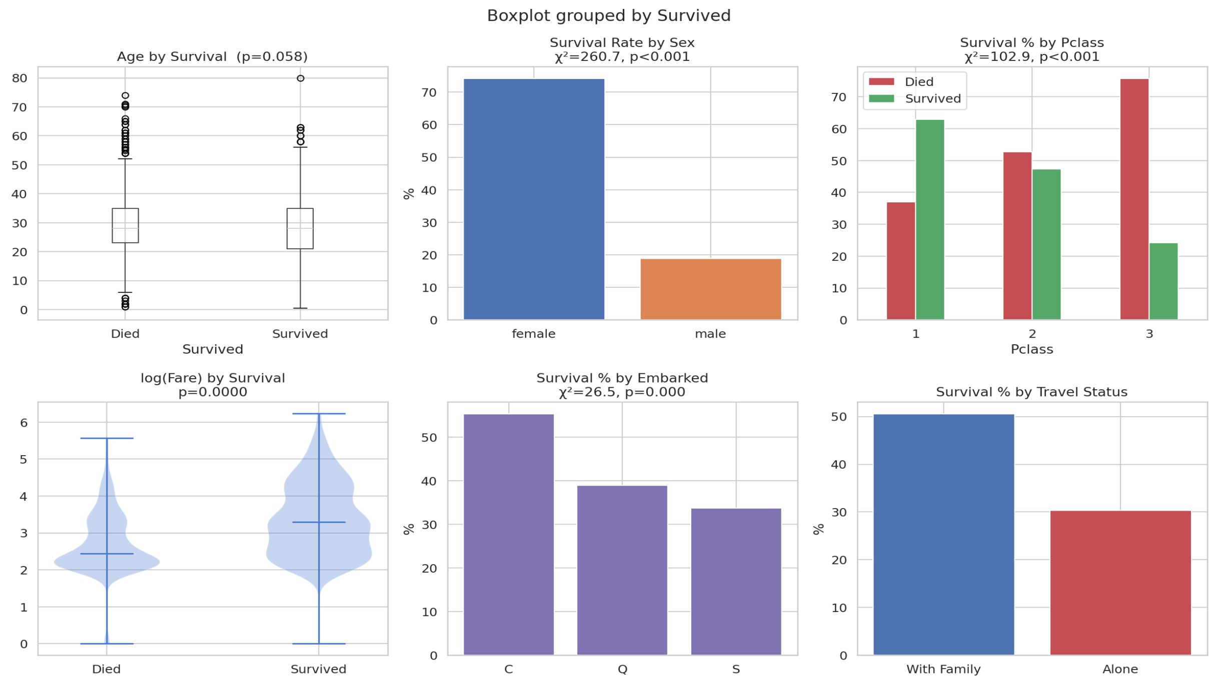

Figure 1.3 — Hypothesis Testing Visualisations

1.4 Correlation vs Causation

Pearson correlations with Survived:

Feature	Pearson r	Spearman rho	Interpretation
log(Fare)	+0.330	+0.324	Higher fare → more likely to survive
Fare	+0.257	+0.324	Same signal; log-scale more linear
Pclass	–0.338	–0.340	Higher class (lower number) → survival
IsAlone	–0.203	–0.203	Travelling alone associated with lower survival
Age	–0.065	–0.037	Weak, not significant

Causation note: Fare correlates with survival (r=0.33), but fare did not save lives. The true causal path: *Social class → Cabin deck → Proximity to lifeboats → Survival probability*. Fare and class are correlated proxies for the same underlying variable — position in the ship's social hierarchy.

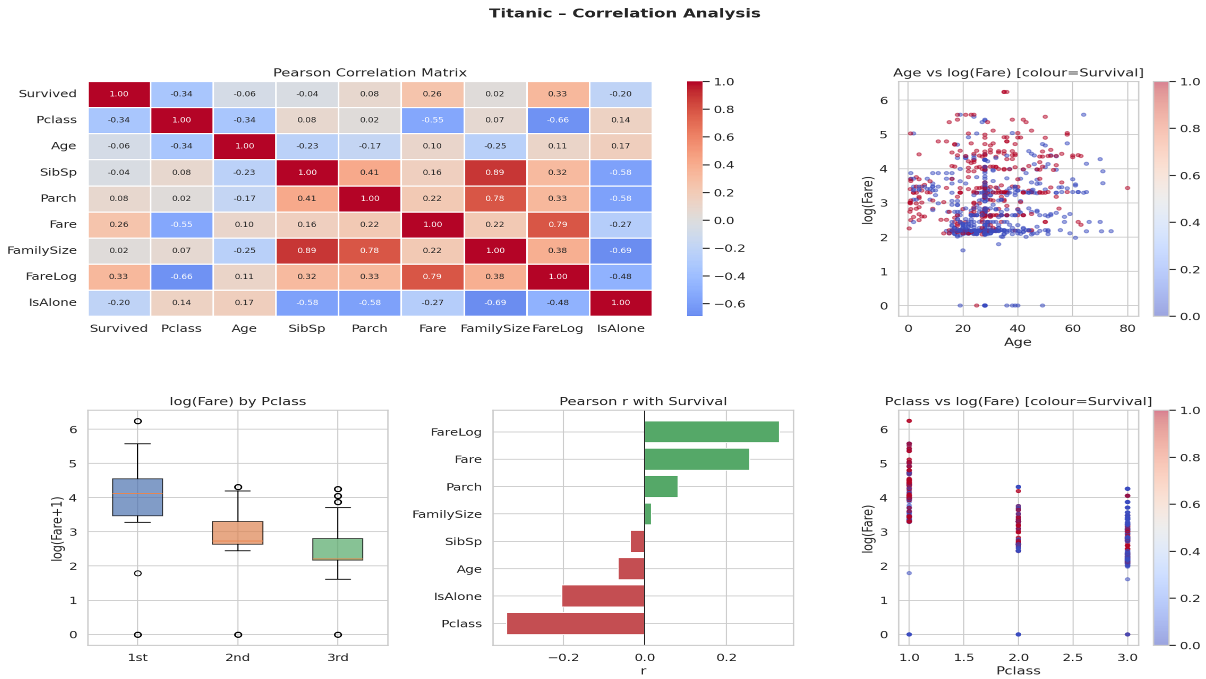

Figure 1.4 — Correlation Analysis & Heatmap

TASK 2: CALIFORNIA HOUSING DATASET

Dataset: 20,640 census block groups, 10 columns. Target: median_house_value. Cleaned: total_bedrooms imputed (207 missing, median). Engineered: rooms_per_household, bedrooms_per_room, population_per_household, log_house_value, log_income.

2.1 Descriptive Statistics

Feature	Mean	Std Dev	Skewness	CV
median_house_value	$206,856	$115,396	0.98	0.558
median_income (x$10k)	3.87	1.90	1.65	0.491
housing_median_age	28.6 yrs	12.6 yrs	0.06	0.439
rooms_per_household	5.43	2.47	20.70*	0.456
bedrooms_per_room	0.213	0.058	2.25	0.272

* rooms_per_household has extreme outlier blocks (one block: 141 rooms/household). This inflates skewness; the bulk of the distribution is clustered at 4–6 rooms.

- **House value** is right-skewed (0.98); log-transform brings skewness near zero.
- **Island properties** average $380k vs Inland $125k (3× premium).
- **Median income** is strongly right-skewed (1.65); high-earning blocks pull the mean above the median.

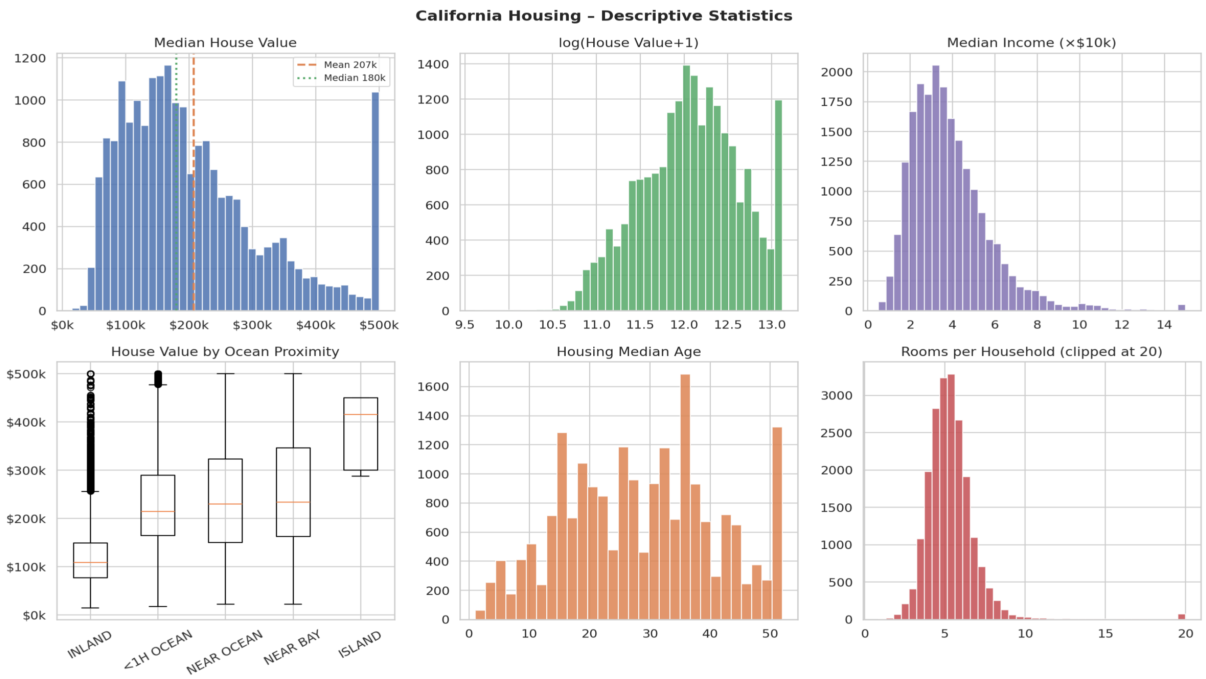

Figure 2.1 — California Housing Descriptive Statistics

2.2 Probability Distributions

None of the raw continuous features are normally distributed (Shapiro-Wilk and D'Agostino tests all $p<0.001$). Log-transforming house value (skew: 0.98 → ~0) and income (skew: 1.65 → ~0.27) substantially improves normality — critical for regression models.

Ocean proximity is a categorical distribution: 44.3% of census tracts are within 1 hour of ocean, 31.7% are inland. Island tracts represent only 0.02% of the data.

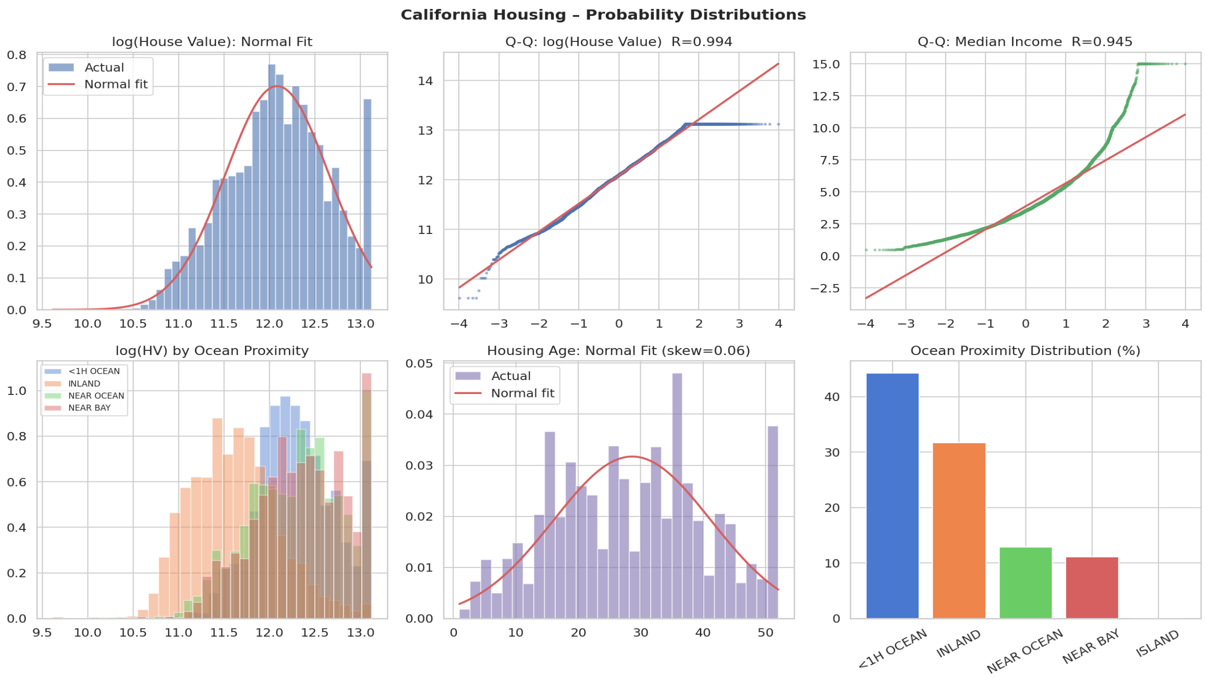

Figure 2.2 — Probability Distributions (fits, Q-Q plots, proximities)

2.3 Hypothesis Testing (alpha = 0.05)

Hypothesis	Test	Statistic	p-value	Decision	Effect
H1: House value equal across all ocean proximity groups	One-Way ANOVA	F=2464	<0.001	Reject H0	Large
Post-hoc: INLAND vs NEAR OCEAN	Mann-Whitney	U=2.95M	<0.001	Reject H0	Large
H2: log(HV) equal for high vs low income blocks	Welch t	t=94.9	<0.001	Reject H0	d=1.32
H3: log(HV) equal across age quartiles	Kruskal-Wallis	H=88.9	<0.001	Reject H0	—
H4: Coastal log(HV) = Inland log(HV)	Welch t	t=97.5	<0.001	Reject H0	d=1.47

Key finding: Every test rejects the null hypothesis. Income group and location (coastal vs inland) both produce large effects (Cohen's d > 1.3). These are not marginal statistical artefacts — they represent meaningful real-world economic differences.

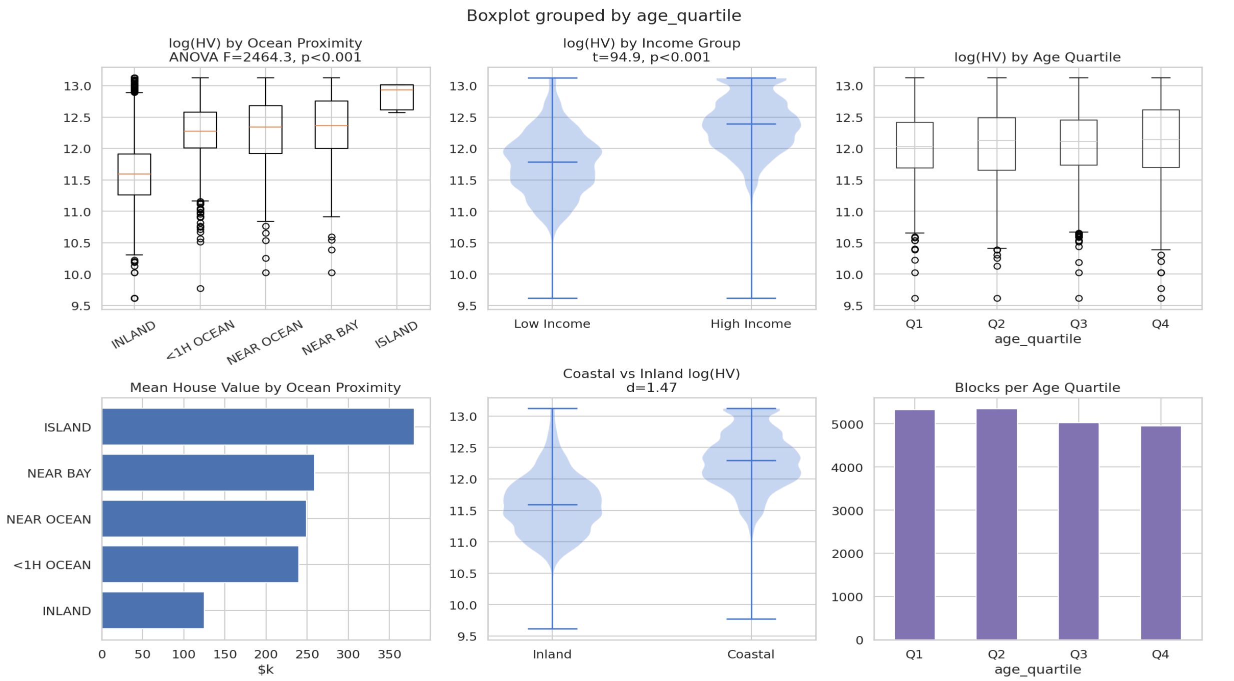

Figure 2.3 — Hypothesis Testing Visualisations

2.4 Correlation vs Causation

Feature	Pearson r	Spearman rho	Causal interpretation
median_income	+0.688	+0.677	Strongest predictor; high-income blocks afford higher prices
rooms_per_hhold	+0.152	+0.263	More spacious units command premium (confounded by income)
housing_median_age	+0.106	+0.075	Older homes in desirable areas (SF, coastal) retain value
latitude	–0.144	–0.166	Geographic proxy; coastal cities cluster at lower latitudes
bedrooms_per_room	–0.256	–0.332	Denser occupancy signals lower-income area, not causation

Spurious correlation example: latitude correlates with income (r=–0.08). This is because high-income areas (Silicon Valley, San Francisco Bay) happen to be at lower latitudes. Latitude does not *cause* income — it is a confounded geographic proxy.

Multicollinearity warning: total_rooms, households, and population are highly correlated with each other (r > 0.9). Using them simultaneously in regression inflates coefficient variance. Engineered ratios (rooms_per_household) are better — they carry the same signal with less redundancy.

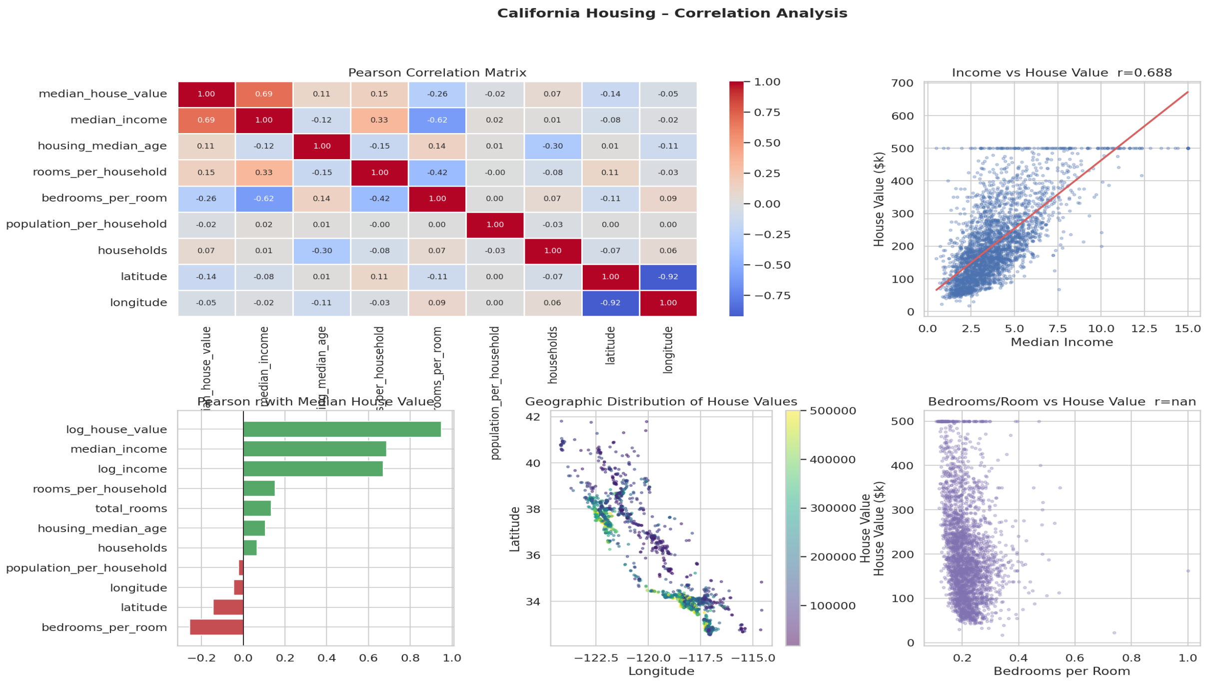

Figure 2.4 — Correlation Matrix & Geographic Value Distribution

COMPARATIVE SYNTHESIS

Cross-dataset statistical lessons from Week 3:

Theme	Titanic	California Housing
Distribution type	Age ≈ Normal; Fare = Log-normal	Income & HV = Log-normal; Age ≈ Normal
Strongest predictor	Sex (Chi2=260.7, V=0.54)	Income (r=0.688, d=1.32)
Log-transform benefit	Fare: skew 4.79 → 0.40	House value: skew 0.98 → ~0.00
Correlation vs Causation	Fare signals class, not direct survival	Income signals affordability, not price-setting
Effect size magnitude	Sex: large; Age: small (d=0.13)	Income/Location: large (d>1.3)
Non-parametric test used	Mann-Whitney U (Age)	Kruskal-Wallis (age quartiles)

Professional LinkedIn Insight

LinkedIn Statistical Insight (for professional share)

"Correlation is not causation — and the Titanic proves it. First-class passengers paid higher fares AND survived at high